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| **Fiscal Year:** | FY 2026 | **Task Last Updated:** | FY 12/18/2025 |
| **PI Name:** | Schuerger, Andrew Ph.D. | | |
| **Project Title:** | Microgravity Can Down-Regulate Host Resistance and thus May Up-Regulate Plant Disease Development in Space | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Plant Biology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research |
| **Start Date:** | 01/01/2022 | **End Date:** | 12/31/2025 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA KSC |
| **Contact Monitor:** | Massa, Gioia | **Contact Phone:** | 321-861-2938 |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 12/31/2025 per PI (Ed., 11/26/24). | | |
| **Key Personnel Changes/Previous PI:** | 1) Dr. Andrew Schuerger (2022-2025 PI) will be retiring on Dec. 31, 2025. 2) Dr. Rob Ferl (current co-I) has agreed to become the 2026 PI for the project to support the science as we work towards a May 12, 2026 launch on Space X-34. 3) Dr. Anna-Lisa Paul (current co-I) will become a 2026 co-PI with Rob Ferl. 4) Kylee Soltez (Masters) will continue on as a co-I on the project. | | |
| **COI Name (Institution):** | Ferl, Robert Ph.D. ( University of Florida, Gainesville )<br>Paul, Anna-Lisa Ph.D. ( University of Florida, Gainesville )<br>Reed, David M.S. ( Redwire, Inc. )<br>Soltez, Kylee ( Dept. of Plant Pathology, Univ. of Florida ) | | |
| **Grant/Contract No.:** | 80NSSC22K0209 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Space-faring nations are utilizing small plant-growth payloads in microgravity (micro-g) to develop the knowledge and technology infrastructures to advance the development of food production systems on other planetary bodies. As the use of small plant-growth payloads in micro-g continues, plant disease outbreaks will increase over time, once the systems are integrated into the open-air microbiomes of spacecraft. This situation presents an opportunity to address directly Section 2.3.2.B of NASA Solicitation 2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research – the combined effects of various space-relevant stressors – in a manner that further enables exploration.<br>A solid literature base exists that indicates that plant host resistance is down-regulated in micro-g and includes studies that describe decreased cell wall rigidity, cell wall thickness, cellulose and matrix polysaccharides, lignin, and altered host-resistance gene pathways in micro-g. An equally solid literature base indicates that microbial virulence may be up-regulated in microgravity and includes up-regulation of virulence in microbe/microbe, microbe/insect, and pathogen/plant interactions.<br><br>However, no data exists on the interactions of a foliar phytopathogen on a plant host with concomitant host-resistance transcriptomics data. The alternative hypothesis (Ha) for the International Space Station (ISS)-flight experiment is: Microgravity Can Down-Regulate Host Resistance and thus May Up-Regulate Plant Disease Development in Space. Results will fill key knowledge gaps into how plant diseases and host resistance are affected by micro-g.<br><br>Proposed here is a novel flight experiment that will study the development of a foliar plant pathogen (i.e., phytopathogen) on the well-studied, Arabidopsis thaliana (At) host. The phytopathogen – Golovinomyces cichoracearum (Gc), a powdery mildew fungus - on A. thaliana is a well-studied pathosystem. The Gc/At pathosystem is chosen here because (i) both Gc and At are sequenced and annotated, (ii) extensive literature is available on host-resistance in At, (iii) diverse At cultivars with differential expression of easily measured host resistance mechanisms are available, (iv) most stages of the Gc life-cycle are on external surfaces of leaves and can be easily observed, (v) the expected ease of sanitizing flight hardware, and (vi) maximized crew safety on the ISS because Gc has no established interaction with humans.<br><br>We will use the Multi-Use Variable-Gravity Platform (MVP) facilities built by Redwire, Inc. (previously Techshot, Inc.) (2 units are in orbit on the ISS) to investigate the development of Gc on leaves of At. Each MVP has two independently controlled centrifuge rotors fitted with up to 4 Phytofuge plant-growth modules that will be rotated at 1g or left stationary in micro-g. Each Phytofuge unit has three separate petri dishes with light-emitting diode (LED) illumination and an internal camera.<br><br>Seed of two cultivars of At will be (1) sown onto growth media in independent petri dishes, (2) held dormant for up to 30 days, and (3) once in orbit, one-half of the petri dishes will be inoculated with the powdery mildew phytopathogen Gc. The aerial mycelia, conidiophores, and spores of Gc will be allowed to develop for 10-12 days and then leaves harvested for two separate research pipelines. First, half of the healthy and half of the infected At plants will be fixed in 3% buffered glutaraldehyde and stored at 4C until processed on the ground for fluorescent staining, Scanning Electron Microscopy (SEM), and Transmission Electron Microscopy (TEM) studies into the process of host infection. Second, the remaining healthy and infected plants will be fixed in RNAlater, frozen at -80C, and later processed for transcriptomics of host-resistance genes.<br><br>Results will inform future horticulturists, space engineers, and technologists of the risks of maintaining plant-host resistance in space when challenged by an airborne phytopathogen. The results will also assist in the design of future plant-growth modules for crewed missions to the Moon and Mars. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Plant disease development in space has profound impact on the future of human exploration of the Solar System. Currently the assumption is that plants grown in space-based bioregenerative life support systems (BLSS) modules will develop normal plant-resistance mechanisms to exposure of biological agents (e.g., bacteria, fungi, viruses). If disease resistance is "normal" in space-based BLSS modules, the use of crops for food, oxygen, and water recycling will be a viable option for crewed habitats on the Moon and Mars. In contrast, if plant diseases develop more quickly in space than on Earth, new and unique plant production protocols may have to be developed. The research outlined in this project seeks to identify if "plant resistance" against a fungal phytopathogen in microgravity progresses normally in the mustard plant, Arabidopsis thaliana. The fungal pathogen has the general name of "powdery mildew", but the species name is Golovinomyces cichoracearum. Powdery mildew phytopathogens have no proven disease risk to humans, and thus, there is no health risk to the astronauts on the International Space Station (ISS) during the flight experiment.<br>The hypothesis being tested here is (Ha): Microgravity can down-regulate plant host resistance, and thus, may up-regulate plant disease development in space.<br><br>The primary benefit to Earth-based agriculture will be to identify how disease resistance mechanisms operate under the unusual conditions of microgravity. Results may identify how to improve disease resistance in field crops on Earth. |
| **Task Progress:** | 2025 Summary of the MVP-Plant-02 ground research.<br>The pathosystem to be tested in the International Space Station (ISS) flight experiment is the fungal powdery mildew (PM) phytopathogen Golovinomyces cichoracearum (Gc) infecting leaves of Arabidopsis thaliana (At). The ISS flight experiment will be conducted in two Multi-Use Variable-g Platforms (MVP) built by Redwire and currently on board the ISS.<br><br>The four major accomplishments for 2025 were:<br><br>a) We successfully completed the Science Verification Test (SVT) between 13-Jan-2025 and 10-Feb-2025. Results indicated that At plants and Gc disease developed nominally within the Phytofuge Experiment Module (PEM) petri dishes. All success criteria were satisfied.<br><br>b) We successfully completed the Experiment Verification Test (EVT) between 13-May-2025 and 10-June-2025. Results confirmed that A. thaliana and G. cichoracearum developed nominally within the PEMs. Although we had a fixation anomaly (see below), the EVT was completed successfully. All success criteria were satisfied.<br><br>c) We completed the Experiment Requirements Document (ERD) in late 2024, and it was updated (as required) for the |

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| | SVT and EVT.<br><br>d) We are working on an advanced draft of the Science Readiness for Flight Review (SRFR) document, and we should complete it before 15-Oct-2025.<br><br>Overall, we have completed all major Gc/At development goals, the SVT, the EVT, and are ready to conduct the flight experiment in 2026. As of this writing, the first flight opportunity for an ISS launch is in April 2026 with the MVP-Plant-02 samples returned in late July 2026. However, we have a funding shortfall, primarily in salaries for the project. A brief annual review and the 2026 Statement of Work (SOW), Budget Narrative, and Budget Details have been submitted as an augmentation request for the Gc/At flight experiment. |
| **Bibliography Type:** | Description: (Last Updated: 01/12/2026) |
| **Articles in Peer-reviewed Journals** | Schuerger AC, Soltez KS, Paul AL, Ferl R. "Simulated microgravity enhances germ tube elongation by Golovinomyces cichoracearum on Cucurbita pepo and Arabidopsis thaliana leaves." Phytopathology. 2025 Dec 15(ja). https://doi.org/10.1094/PHYTO-11-25-0359-R ; PMID: 41397306 , Dec-2025 |